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Method for producing a cemented carbide body

Abstract

The invention relates to a method for producing a cemented carbide body, wherein in a mixing and/or grinding process, preferably in a wet grinding process, tungsten carbide powder (WC powder) and metallic binder material comprising cobalt powder (Co), nickel (Ni) and aluminum (Al) are mixed to form a powder mixture, wherein a green compact is pressed from at least a portion of the powder mixture and wherein the green compact is sintered in a sintering step under the influence of temperature and pressure in such a way that the cemented carbide body is formed after a cooling step following the sintering step. To create an easily controllable and reliable process for the production of a cemented carbide body, which is characterized by improved wear resistance and, at the same time, a high fracture strength, provision is made in accordance with the invention for nickel aluminide, preferably nickel aluminide powder, in particular Ni.sub.3Al powder, to be added to the mixing and/or grinding process as an intermetallic phase material.

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Background/Summary

[0001] The invention relates to a method for producing a cemented carbide body, wherein in a mixing and/or grinding process, preferably in a wet grinding process, tungsten carbide powder (WC powder) and metallic binder material comprising cobalt powder (Co), nickel (Ni) and aluminum (Al) are mixed to form a powder mixture, wherein a green compact is pressed from at least a portion of the powder mixture and wherein the green compact is sintered in a sintering step under the influence of temperature and pressure in such a way that the cemented carbide body is formed after a cooling step following the sintering step.
[0002] EP 2 691 198 B1 describes a cemented carbide material, namely a hard metal body (tungsten carbide material), and a method for its production. According to this known method, a powder comprising coarse-grained tungsten carbide, a superstoichiometric proportion of carbon and cobalt powder, is mixed. In addition, powdered tungsten was added to the powder. The tungsten powder and the cobalt powder had a mean particle size of approx. 1 μm. The coarse grain tungsten carbide had a mean particle size of 40.8 μm.
[0003] Then this powder was ground in a ball mill and hexane and paraffin wax were added. A green compact was pressed from this mixture and subsequently this green compact was sintered. After the sintering process, the obtained cemented carbide material was subjected to a heat treatment. It was heated to 600° C. and kept at this temperature for 10 hours.
[0004] After a subsequent cooling process, the cemented carbide material was analyzed. It turned out that there are nanoparticles in the binder phase of the cemented carbide material, wherein the nanoparticles have a size smaller than 10 nm. The nanoparticles were formed by the Eta phase (Co.sub.3W.sub.3C) or (Co.sub.3W.sub.6C) or the Theta phase (Co.sub.2W.sub.4C). The particle size of the nanoparticles was smaller than 10 nm.
[0005] It has been shown that nanoparticles are accompanied by a reinforcement of the binder phase. This can increase the hardness of the cemented carbide material. A disadvantage of these materials is the lack of thermal stability of the nanoparticles. As a result, they are only suitable to a limited extent for high-temperature applications or for applications, in which a high temperature input occurs.
[0006] During rock machining and asphalt and concrete milling, friction generates very high temperatures on the tool surface. The hard material tungsten carbide has a high hot hardness at these temperatures and is not much affected by them. However, the strength of the metallic binder drops dramatically at these temperatures. The reduced strength of the metallic binder results in increased abrasive wear and or extrusion of the binder phase as a result of the stresses imposed by the application. As a result, the hard metal can no longer hold the tungsten carbide grains.

[0007] The invention addresses the problem of providing an easily controllable and reliable method for the production of a cemented carbide body, which is characterized by improved wear resistance and, at the same time, high fracture strength.
[0008] This problem is solved by adding nickel aluminide, preferably nickel aluminide powder, in particular Ni.sub.3Al powder, as intermetallic phase material to the mixing and/or grinding process.
[0009] Nickel aluminide, in terms of the invention, may be an intermetallic phase material comprising at least Ni and Al, wherein Ni and Al are bonded to each other in a crystal structure.
[0010] In terms of the invention, nickel aluminide shall be understood as to apply to alloyed nickel aluminide (e.g., tin alloyed nickel aluminide).
[0011] For the purposes of the invention, nickel aluminide can be understood to denote an amorphous material, in particular in powder form.
[0012] Nickel aluminide can be added to the mixing or grinding process without any special precautions having to be taken in terms of occupational safety or health. In particular, no deoxidation is required, which would be necessary if elemental Al were added. In addition, this results in the option of the nickel aluminide being easily added to the mixing and/or grinding process in precise doses, thus permitting a reliable and reproducible production in a simple manner.
[0013] The method according to the invention can be used to produce a cemented carbide material, in particular a hard metal, which then either already has a reinforced binder phase and/or is prepared to form a reinforced binder phase. In both cases, the binder phase is reinforced via an intermetallic phase material that forms including the Ni and Al of the added nickel aluminide.
[0014] According to a preferred embodiment of the invention, provision may be made for the cooling step following the sintering step and/or a thermal treatment of the cemented carbide body to be controlled such that an intermetallic phase material is formed in a binder phase of the cemented carbide body, wherein at least a portion of the intermetallic phase material is preferably formed according to the structural formula (M,Y).sub.3(Al,X) wherein M=Ni, Y=Co and/or another constituent, and X=tungsten and/or another constituent.
[0015] Preferably provision may be made, during the cooling step, to keep the sintered body in a temperature range from 400° C. to the solvus temperature of the sintered body for 0.25 to 24 hours. There, the intermetallic phase material forms reliably and in sufficient quantity and size to achieve an effective reinforcement of the binder phase.
[0016] If the intermetallic phase material is already formed in the cooling step and is then present in the binder phase, it directly reinforces the latter. This results in a particularly simple design of the method.
[0017] If the dissolved elements Ni and Al are present in the binder phase in the cemented carbide material after the sintering process, thermal treatment of the cemented carbide material can form intermetallic phase material, which then results in a reinforcement of the binder phase and the desired improved wear resistance.
[0018] For instance, during this thermal treatment, at least a portion of the elements Ni, Co, W and Al from the binder phase combine to form intermetallic phase material. At least a portion of this intermetallic phase material can be formed according to the structural formula (M,Y).sub.3(Al,X), wherein M=Ni, Y=Co and/or another constituent, and X=tungsten and/or another constituent.
[0019] The thermal treatment of the cemented carbide material can be performed in different ways, which are suitable for forming the intermetallic phase material as intended.
[0020] In particular, this may be a thermal treatment, in particular an external heat input into the cemented carbide material. The thermal heat treatment can be effected, for instance, by an active source providing heating or cooling and in that this source introduces heat into or extracts heat from the material. For instance, the formation of the intermetallic phase material can be performed in a furnace, into which the cemented carbide material has been introduced. It is also conceivable that at least part of the surface of the cemented carbide material is acted upon by a heating device, for instance a burner.

[0021] Alternatively, it is conceivable that an excitation source is present that introduces energy into the cemented carbide material to generate heat therein. This can be, for instance, an induction coil or a laser device.
[0022] Alternatively, it is also conceivable that the heat is generated by passive heating, i.e., by using the cemented carbide material in an operating state, preferably as intended, or in the course of a method step in which the cemented carbide material is processed, in particular installed.
[0023] For instance, heat is generated in the cemented carbide material by friction, such as occurs during the intended use, in particular the intended application, in particular the application of tools, of the cemented carbide material. If it is a tool, it is moved relative to the object to be machined (for instance, in the case of a road milling pick, this road milling pick is moved relative to a road surface), frictional energy is generated, which results in heat generation in the cemented carbide material. The heat generated in this process can be used to achieve a self-reinforcing effect of the binder phase because of the, at least partial, formation of intermetallic phase material in the cemented carbide material.
[0024] An intended operating condition can also be understood as the operational use of the cemented carbide material at an operating temperature suitable to form intermetallic phase material.
[0025] It is also conceivable that the cemented carbide material is passively heated in the course of a method step, in which the cemented carbide material is applied to a holder, for instance to a tool base body or tool head. The heat generated can be used to form the intermetallic phase material. One conceivable joining process is a welding process, for instance a friction welding process, an electron beam welding process, a brazing and soldering process, for instance a brazing process, a furnace brazing process, an inductive brazing process, a diffusion brazing process, a plating process, for instance explosive plating.
[0026] The intermetallic phase material forms a crystalline intercalation in the metallic binder.
[0027] This intermetallic phase material has significantly higher strength compared to the metallic binder material, in which it is intercalated, in particular at higher temperatures. At the surface of the cemented carbide material exposed to the wear attack, the intermetallic phase material reduces erosion or extrusion of the metallic binder material when it is used, for instance, in a soil cultivation tool.
[0028] The motion of the soil cultivation tool and the loosened soil material and the remaining soil material causes an abrasive and mechanical stress on the cemented carbide material. The tungsten carbide grains in the cemented carbide material provide sufficient wear resistance against this wear attack. The problem in the state of the art is the binder material, which has a significantly lower strength than the tungsten carbide. Because the intermetallic phase material is now integrated or forms in the binder phase, any rapid erosion or extrusion of the metallic binder material is prevented.
[0029] Moreover, surprisingly, the intermetallic phase material has also been shown to be able to reinforce the internal structure of the cemented carbide material. If strong impact stresses occur, the crystals of the intermetallic phase material reduce or prevent any sliding of the tungsten carbide grains in the area of the interconnecting binder phase and thus reduce or prevent any excessive plastic deformation of the binder phase. In particular, the individual crystals of the intermetallic phase material prop each other. This has a considerable advantage, in particular at high tool application temperatures. For instance, when Co is used in the binder phase, the strength of the Co in the binder phase is reduced at such temperatures, but the intermetallic phase material still reliably provides sufficient support for the binder material.
[0030] Overall, it has been shown that a significant increase in the wear resistance of the cemented carbide material can be achieved based on the method according to the invention. Tests have shown that, for instance, the use of the cemented carbide material in the form of a pick tip of a round pick for road-milling machines results in an increase of wear resistance of up to 50%! It has been shown

that such a significant increase in wear resistance can be achieved when milling road surfaces, both asphalt and concrete.
[0031] The cemented carbide material can be used in particular to design the working areas of tools for working, loosening, conveying and processing plant-based or mineral materials or building materials, in particular in the areas of agriculture or forestry or road construction, mining or tunnel construction.
[0032] If intermetallic phase material is present in the cemented carbide material, according to one variant of the invention provision may be made for the proportion of metallic binder material in the cemented carbide material to be 1 to 28 wt. %, preferably 1 to 19 wt %. In so doing, apart from unavoidable impurities, all or virtually all of this metallic binder material may be formed by Co. Such a choice of material results in a particularly tough binder phase, which can be effectively reinforced by the existing or the forming intermetallic phase material.
[0033] According to one embodiment of the invention, provision may be made for the sum of the elements Ni and Al in the cemented carbide material to be 1 to 28 wt %, preferably 1.5 to 19 wt %, at least in one segment of the cemented carbide body. These range specifications take into account Ni and Al both from any intermetallic phase material present and dissolved Ni or Al in the binder phase. Such compositions can be used to create in particular sophisticated hard-metal tools for soil cultivation.
[0034] According to a possible variant of the invention, in order to be able to manufacture such a cemented carbide material, provision may be made for the green compact to contain 70 to 95 wt %, preferably 80 to 95 wt %, tungsten carbide (WC), 1 to 28 wt %, preferably 1 to 19 wt %, cobalt (Co) and 1 to 28 wt %, preferably 1.5 to 19 wt %, nickel aluminide, preferably as an intermetallic phase material.
[0035] If no or only little intermetallic phase material is present in the cemented carbide material, according to a variant of the invention provision may be made for the proportion of binder phase in the cemented carbide material to be 5 to 30 wt %, preferably 5 to 20 wt %. In this case, half or most of the binder phase may be formed by the metallic binder material Co. In addition, Al and Ni may be dissolved as metallic binder material in the binder phase. Finally, other elements and unavoidable impurities may be present in the binder phase.
[0036] According to the invention, in addition to unavoidable impurities, provision may also be made for the binder phase to contain other constituents besides Co, in particular dissolved W, C, Ni, Al and/or Fe.
[0037] According to the invention, the intermetallic phase material optionally present in the binder phase may be formed according to the structural formula (M,Y).sub.3(Al,X), or the intermetallic phase material may be formed according to this structural formula, based on the elemental composition in the final cemented carbide material, wherein M=Ni, Y=Co and/or another constituent, and X=tungsten and/or another constituent.
[0038] If there is intermetallic phase material present in the binder phase, preferably at least for the majority of the crystals of the intermetallic phase material Y=Co and X=W.
[0039] Accordingly, the composition of the dissolved constituents in the binder phase in the cemented carbide material is such that the intermetallic phase material can be formed by heat treatment or heat exposure in this way (see above).
[0040] Additionally, for some or all of the crystallites of the intermetallic phase material (Al, X) may be present such that X is present in the form of both W and Mo and/or Nb and/or Ti and/or Ta and/or Cr and/or V. Accordingly, the composition of the dissolved constituents in the binder phase in the finished cemented carbide material can be selected to permit the intermetallic phase material to form due to heat treatment or heat exposure in this manner (see above).
[0041] According to the invention, provision may be made for the binder phase to comprise two or more intermetallic phase materials or only one intermetallic phase material and/or that the cemented carbide material is prepared such that thermal treatment or thermal exposure causes two

or more intermetallic phase materials or only one intermetallic phase material to be formed.

[0042] According to a preferred embodiment of the invention, provision may be made for the binder phase to have the chemical element composition listed below:

[0043] Ni>25 wt %, Al>4 wt %, the balance is made up of Co and dissolved binder constituents, for instance W and/or C,

preferably Ni>35 wt %, Al>5 wt %, the balance is made up of Co and dissolved binder constituents, for instance W and/or C,

particularly preferably Ni>40 wt %, Al>6.5 wt %, the balance is made up of Co and dissolved binder constituents, for instance W and/or C.

[0044] These figures refer to the total content of the individual substances in the binder phase. The data thus take into account the individual elements in dissolved form and/or additionally also the respective elements as they are bound in the intermetallic phase material.

[0045] It has been shown that an effective reinforcing effect for the binder phase occurs at the specified values.

[0046] To be able to effect an optimum reinforcement of the binder phase using the intermetallic phase material, provision may be made in the context of the invention for the ratio of the mass fractions Al to Ni to be >0.10, preferably >0.12.

[0047] Preferably, provision may also be made in this context for the ratio of the mass fractions Al to Ni to be ≤0.46, preferably ≤0.18, more preferably ≤0.16.

[0048] The above data, which relate the ratio of mass fractions Al to Ni, take into account the total mass fraction, i.e., both the dissolved elements Al and Ni and Al and Ni in the intermetallic phase material (if present). Testing for the mass fractions of Al and Ni can be performed by means of a common ICP measurement.

[0049] One conceivable variant of the invention is such that the cemented carbide material has at least two volume segments, wherein the relative proportion of intermetallic phase material, based on a unit volume, is greater in the first volume segment than in the second volume segment. By designing at least two volume segments, it is possible to specifically influence the properties, in particular the tool properties, of the cemented carbide material. For instance, a zone subject to high abrasive wear may comprise the volume segment having a higher relative proportion of intermetallic phase material. In contrast, a zone that has to meet special toughness requirements can comprise the volume segment having a lower relative proportion of intermetallic phase material. For the sake of completeness, it should be noted at this point that the second volume segment, which has a lower relative proportion of intermetallic phase material compared to the first volume segment, that this second volume segment may also be without any intermetallic phase material.

[0050] A conceivable alternative invention is such that the first volume segment, which has the high relative proportion of intermetallic phase material, is delimited by at least one segment of the surface of the cemented carbide material. In this way, a high resistance to abrasion is achieved on the surface of the cemented carbide material there. Preferably, the second volume segment is not located adjacent to a surface of the cemented carbide material, but is located inside the cemented carbide material. Here, it provided high fracture stability.

[0051] Alternatively, provision may also be made for the second volume segment, which has a lower relative proportion of intermetallic phase material or no intermetallic phase material compared to the first volume segment, to be delimited by at least one segment of the surface of the cemented carbide material and preferably the first volume segment is not adjacent to a surface of the cemented carbide material. In this way, a tool that optimally adapts its cutting contour to the cutting task during tool application to achieve a good tool life can be created. In particular, it can be used to implement a so-called re-sharpening effect.

[0052] One conceivable variant of the invention is such that the binder phase, in particular the metallic binder material and/or the intermetallic phase material comprise Nb and/or Ti and/or Ta, and/or Mo and/or V and/or Cr, wherein preferably one or more of these materials is/are present dissolved in the binder phase and/or as carbides. In that way, an increase of the solvus temperature and the strength of the existing and/or the intermetallic phase formed by thermal action or treatment can be achieved. As a result, less intermetallic phase material is required while maintaining the strength of the cemented carbide material. Or the binder strength and thus the heat resistance increases due to the addition.

[0053] However, it is also conceivable that one or more of the aforementioned constituents is/are integrated into the crystal lattice of at least part of the intermetallic phase material or can be integrated into the crystal lattice of the intermetallic phase material by the thermal treatment (heat treatment or heat exposure). For instance, the titanium atom (or another material of the aforementioned group) primarily occupies the lattice site of Al or W in the crystal lattice of the intermetallic phase material and, like W, increases the precipitation temperature for the intermetallic phase material.

[0054] This means that, if present in the cemented carbide material, on the one hand, the intermetallic phase material can be precipitated more effectively during sintering, because precipitation starts at higher temperatures, because the diffusion rate is significantly higher there.

[0055] On the other hand, this measure achieves a high heat resistance, since, as mentioned, the solvus temperature of the cemented carbide material is increased. In other words, the temperature required to redissolve the intermetallic phase material in the cemented carbide material increases.

[0056] According to the invention, provision may be made for the proportion of Mo and/or Nb and/or Ti and/or Ta and/or Cr and/or V in the binder phase to be ≤15 at %. In principle, the above-mentioned elements do form carbides. In the context of the invention, provision may now be made for the material composition to be chosen such that small amounts of these elements, according to the solubility product and their affinity to carbon, are dissolved in the binder phase, i.e., they can thus be incorporated into the crystal lattice of the intermetallic phase material and/or be dissolved in the metallic binder phase. If a cemented carbide material is desired that has high toughness of the binder phase, then the carbide fraction should be kept small. The sum of these materials should then be present at a proportion ≤15 at %.

[0057] Furthermore, advantageously the powder mixture (in particular the powder mixture of the green compact) for the production of the cemented carbide material can be set stoichiometrically with regard to the carbon content, because the titanium (and/or Mo and/or Nb and/or Ti and/or Ta and/or Cr and/or V) takes over the role of the tungsten.

[0058] According to one design variant of the invention, provision may be made for the carbon content to be set stoichiometrically or also sub-stoichiometrically in the powder mixture (in particular the powder mixture of the green compact). This measure prevents or minimizes graphite precipitation in the sintered material due to overstoichiometric carbon content. The inventors have recognized that such intercalations have a detrimental effect on the fracture strength of the cemented carbide material.

[0059] According to the invention, in particular provision may be made for the carbon content in the cemented carbide material to be in the range from: [0060] C.sub.stoich (wt %) −0.003 times the binder content (wt %) to C.sub.stoich (wt %) −0.012 time the binder content wt %,

to be preferably in the range from: [0061] C.sub.stoich (wt %) −0.005 times the binder content (wt %) to C.sub.stoich (wt %) −0.01 times the binder content wt %.

[0062] In the context of the invention, the advantageous effects described above are particularly pronounced in the case of coarse-grained hard metal. In a preferred embodiment of the invention, provision may therefore be made for the dispersed tungsten carbide to be present in the cemented carbide material as grains having a mean particle diameter, measured according to DIN ISO 4499 Part 2, in the range from 1 to 15 μm, preferably in the range from 1.3 to 10 μm, particularly preferably in the range from 2.5 to 6 μm.

[0063] Preferably, provision is made for the maximum content of Fe in the binder phase or in the green compact to be 5 wt % and/or for other unavoidable impurities to be present in the binder

material.
[0064] If provision is made for the existing intermetallic phase (M,Y).sub.3(Al,X) or the intermetallic phase resulting from heat treatment or heat exposure to have a crystal structure L12 (space group 221) according present to ICSD (Inorganic Crystal Structure Database), then a microstructure in the binder phase results, in which the crystals of the intermetallic phase can effectively prop each other in the metallic binder material when the cemented carbide body is subjected to heavy loads.
[0065] Preferably, for the intended use, preferably of ground machining tools, provision is made for the intermetallic phase material present or resulting from heat treatment or heat exposure to have a maximum size of 1500 nm, preferably a maximum size of 1000 nm.
[0066] According to a preferred embodiment of the invention, provision may be made for the cemented carbide material to be free or as free as possible from the Eta phase and/or Al.sub.2O.sub.3. The inventors have recognized that the maximum proportion of the Eta phase or the maximum proportion of Al.sub.2O.sub.3 should not exceed 0.6 vol % based on the total cemented carbide material. If both substances are present in the cemented carbide material, it is advantageous if the total of the Eta-phase material and of Al.sub.2O.sub.3 is at most 0.6 vol %.
[0067] The particle size of Al.sub.2O.sub.3 and/or of the Eta-phase material is advantageously at most 5 times the mean WC grain size, wherein the mean WC grain size and the particle size of Al.sub.2O.sub.3 and/or of the Eta phase material can be determined using the linear-intercept technique, according to DIN ISO 4499 Part 2.
[0068] The toughness of the cemented carbide material can be negatively affected by the Eta phase or Al.sub.2O.sub.3. At higher Eta-phase contents, the cemented carbide material is only of limited suitability for use in demanding soil cultivation tools. The same applies to Al.sub.2O.sub.3.
[0069] The above mentioned problem of the invention is also solved using a method for manufacturing a tool, in particular a crushing tool, a soil cultivation tool, preferably for a road milling machine, a recycler, a stabilizer, an agricultural or silvicultural soil cultivation machine, having a base body, which comprises a working area, wherein at least one working element, consisting of a cemented carbide material, produced according to any of claims **1** to **22**, is kept on the working area, preferably by means of a material bond, in particular a soldered or brazed joint, in particular a brazed joint.
[0070] Preferably, the cemented carbide material in the working area forms a cutting body having a cutting tip or a blade or a cutting edge or a working edge. It is also conceivable that the cemented carbide material is a hardfacing.
[0071] As mentioned above, according to the invention provision may be made accordingly for the working element to be in the form of a cutting element, preferably having at least one cutting edge and/or at least one cutting tip, or in the form of a wear protection element, in particular a protective plate, a protective strip, a protective pin, a protective projection or a protective stud.
[0072] A particularly preferred application of the invention results when provision is made for the tool to be a cutting tool, a milling pick, in particular a road milling pick or mining milling pick, a plowshare, a cultivator tip, a drilling tool, in particular a soil auger, a crushing tool, for instance a crushing bit or a crushing bar, a mulching tool, a wood chipping tool or a shredding tool, a fractionation tool, for instance a screen.
[0073] A further particularly preferred application of the invention is such that the milling pick comprises a pick head and a pick shank connected directly or indirectly thereto, and that the working element is held on the pick head.
[0074] For instance, provision may also be made for the working element to be formed by the cemented carbide material according to the invention, wherein this working element forms a support for a superhard cutting tip consisting, for instance, of PCD material.
[0075] As described above, the cemented carbide material may be a hard metal having a reinforced binder phase. This reinforcement may occur by precipitation of intermetallic phase material during

cooling in the sintering process and/or it is such that the intermetallic phase material is formed in a thermal process subsequent to the sintering process in which the cemented carbide material is brought to a temperature that allows precipitation of the intermetallic phase material in the cemented carbide material.

[0076] A nominal composition at the weighing of the raw materials of 70 to 95 wt % WC, 1 to 28 wt % metallic binder and 1 to 28 wt % nickel aluminide (e.g., intermetallic phase) can selected for the production of a hard metal according to the invention. The metallic binder may have the elements Co, optionally Fe and/or other constituents. The intermetallic phase at weighing preferably is Ni.sub.3Al.

[0077] The Problem of the invention is also solved using a method for creating a cemented carbide material, wherein first in a first process step a precursor cemented carbide material, in particular a hard metal, is created, containing 70 to 95 wt %, preferably 80 to 95 wt %, tungsten carbide in dispersed form, and a binder phase, wherein the binder phase comprises metallic binder material, wherein the metallic binder material comprises Co, wherein the binder phase comprises the dissolved elements Ni and Al, wherein the binder phase has the chemical element composition listed below:

[0078] Ni>25 wt %, Al>4 wt %, the balance is made up of Co and dissolved binder constituents, for instance W and/or C,

preferably Ni>35 wt %, Al>5 wt %, the balance is made up of Co and dissolved binder constituents, for instance W and/or C,

particularly preferably Ni>40 wt %, Al>6.5 wt %, the balance is made up of Co and dissolved binder constituents, for instance W and/or C,

wherein in a further method step the precursor cemented carbide material is subjected to a heat treatment to form a cemented carbide material comprising intermetallic phase material in the binder phase, wherein the intermetallic phase material is formed at least in part according to the structural formula (M,Y).sub.3(Al,X), wherein M=Ni, Y=Co and/or another constituent, and X=tungsten and/or another constituent.

[0079] The thermal treatment may comprise at least one heating step or at least one cooling step.

[0080] In accordance with the invention, provision may preferably be made for at least portion of the crystals of the intermetallic phase material (M,Y).sub.3(Al,X) to have the crystal structure L12 (space group 221) in accordance with ICSD (Inorganic Crystal Structure Database) after heat treatment.

[0081] In order to be able to achieve a particularly effective reinforcement of the binder phase, in particular in the case of coarse grain hard metal as cemented carbide material, provision may advantageously be made for at least a portion of the intermetallic phase material to have a maximum size of 1500 nm, preferably a maximum size of 1000 nm (measured according to linear-intercept technique using a micrograph).

[0082] Advantageously, provision may be made for the coercivity H.sub.cM of at least one segment, of the cemented carbide material produced according to the method of the invention, preferably of a segment in which intermetallic phase material is present, to be:

H.sub.cM [kA/m]>(1.5+0.04*B)+(12.5−0.5*B)/D+4 [kA/m],

preferably H.sub.cM [kA/m]>(1.5+0.04*B)+(12.5−0.5*B)/D+6 [kA/m],

particularly preferably H.sub.cM [kA/m]>(1.5+0.04*B)+(12.5−0.5*B)/D+10 [kA/m],

wherein B is the proportion of the binder phase in the cemented carbide material in wt % and D is the grain size of the dispersed WC determined by the linear-intercept technique according to DIN ISO 4499, Part 2.

[0083] For common hard metals, with Co in the binder phase and without intermetallic phase material, coercivity is usually used to indirectly determine the mean grain size of the WC for a given binder content. According to the invention, the intermetallic phase material causes a significant increase in coercivity. Thus, the coercivity can be indirectly evaluated as a measure of the reinforcement of the binder phase due to the intercalated intermetallic phase material. The higher the coercivity, the greater the total interface between metallic binder material, intermetallic phase material and WC. A high degree of precipitated intermetallic phase material results in the individual crystals of the intermetallic phase material propping each other well in the binder phase, in particular at high temperatures (in particular at high tool temperatures).

[0084] Coercivity of at least one segment of the cemented carbide material H.sub.cM [kA/m]>(1.5+0.04*B)+(12.5−0.5*B)/D+4 [kA/m] can be used primarily for the above-mentioned wear protection applications, for instance for hardfacing.

[0085] Coercivity of at least one segment of the cemented carbide material, preferably H.sub.cM [kA/m]>(1.5+0.04*B)+(12.5−0.5*B)/D+6 [kA/m], can be used primarily for the above-mentioned demanding soil cultivation tools.

[0086] Coercivities of at least one segment of the cemented carbide material preferably H.sub.cM [kA/m]>(1.5+0.04*B)+(12.5−0.5*B)/D+10 [kA/m] can be used primarily for the high-performance tools mentioned above.

[0087] According to one design variant of the invention, provision may also be made for the coercivity of at least one segment of the cemented carbide material to be 20% higher than the coercivity of a hard metal body having the same composition and WC grain size as the cemented carbide material, wherein the binder phase is formed solely of metallic Co binder; however, the hard metal body does not contain any intermetallic phase material.

[0088] A hard metal body having the same composition is thus a hard metal body, containing 70 to 95 wt % tungsten carbide in dispersed form, and a binder phase, wherein the binder phase comprises metallic binder material without intermetallic phase material, wherein the proportion of metallic binder material in the cemented carbide material is 5 to 30 wt % and apart from that the binder material has the same or approximately the same composition as the binder material of the cemented carbide material according to the invention.

[0089] As was mentioned above, the coercivity indirectly provides an indication of the proportion of intermetallic phase material in the binder phase. Thus, the coercivity indirectly indicates the degree of reinforcement of the binder phase.

[0090] In the context of the invention, the cemented carbide material may be such that the hot compressive strength of the cemented carbide material manufactured in accordance with the method of the invention at a temperature of 800° C. and a strain rate of 0.001 [1/s] is ≥1650 [MPa] and/or that the hot compressive strength of the cemented carbide material at a temperature of 800° C. and a strain rate of 0.01 [1/s] is ≥1600 [MPa] (measurement for a cylindrical specimen having diameter of 8 mm and height of 12 mm). For such a cemented carbide material it is possible to produce, in particular, cutting tips for road milling picks, in which the proportion of the binder phase is 5 to 7 wt. % and the proportion of WC is in the range from 93 to 95 wt %, wherein preferably WC is present as coarse grains having a mean particle size in the range from 2 to 5 μm.

[0091] A particularly preferred variant of the invention provides that the nickel aluminide, preferably the Ni.sub.3Al powder, is produced in a smelting metallurgical process, and/or that the nickel aluminide, preferably the Ni.sub.3Al powder, is added to the mixing and/or grinding process as a material produced in a melt metallurgical process.

[0092] Particularly good processability of the powder mixture in the mixing and/or grinding process results if provision is made for the nickel aluminide, preferably the Ni.sub.3Al powder, to be added to the process having a mean particle size FSSS<70 μm, preferably having a mean particle size FSSS<45 μm.

[0093] According to a variant of the invention, in a preparation step, preferably in a first grinding step, provision may be made for the nickel aluminide to be mixed with grinding liquid and coarse-grained tungsten carbide, preferably having a mean particle size FSSS>20 μm, particularly preferably having a mean particle size FSSS in the range from 30 μm to 60 μm, for instance in the form of macrocrystalline and/or monocrystalline tungsten carbide, in such a way that crushed

nickel aluminide, preferably crushed nickel aluminide powder, in particular crushed Ni.sub.3Al powder, is formed from the nickel aluminide. In this way, an effective comminution of the nickel aluminide is achieved.
[0094] For instance, process control can be such that in the preparation step and/or in a subsequent grinding step pressing aids, at least one alloying constituent and/or cobalt powder is/are added and mixed with the nickel aluminide and/or the crushed nickel aluminide.
[0095] To achieve a particularly good comminution, advantageously provision is made for the nickel aluminide content in the grinding mixture of the preparation step to be in the range from 8 to 50 wt %, preferably in the range from 9 to 25 wt %.
[0096] Particularly preferably, provision may be made in a subsequent grinding step for WC powder to be added to the pre-grinding from the preparation step in such a way that the proportion of WC powder in the grinding mixture obtained is in the range from 70 to 95 wt % and that in the subsequent grinding step the crushed nickel aluminide is ground to finely crushed nickel aluminide.
[0097] Advantageously, the finely crushed nickel aluminide results in a more homogeneous distribution of the intermetallic phases in the sintered body. Furthermore, the WC structure, in particular the WC coarse grain structure, is better preserved in the WC powder due to the shorter grinding process for the added WC.
[0098] Preferably, according to one embodiment of the invention, the green compact is sintered using a liquid phase sintering process in a furnace at a sintering temperature in the range from 1350° C. to 1550° C.
[0099] A particularly effective reinforcement of the binder phase can be achieved if provision is made for at least a portion of the intermetallic phase material in the binder phase to have a maximum size of 1500 nm, preferably a maximum size of 1000 nm (measured in accordance with the linear intercept technique using a micrograph) and/or if provision is made for at least a portion of the crystals of the intermetallic phase material (M,Y).sub.3(Al,X) in the binder phase to have a crystal structure L12 (space group 221) in accordance with ICSD (Inorganic Crystal Structure Database).
[0100] One conceivable process variant is such that atomized nickel aluminide, preferably atomized Ni.sub.3Al powder, is processed as intermetallic phase material in the grinding and/or mixing process. This material is easy to handle. This material can be crushed with little effort to obtain the desired fine grain structure.
[0101] Provision may preferably be made for the mixing and/or milling process to be a multi-stage process comprising at least two mixing and/or milling steps, wherein preferably the nickel aluminide is added before the last milling and/or mixing step.

PRODUCTION (WITH DESCRIPTION OF MEASUREMENT METHODS)

Production

[0102] The production method by which a cemented carbide material containing intermetallic phase material in the binder phase can be produced via a powder metallurgy process routine is described below. The latter is divided into the process steps of producing a compressible powder mixture, shaping, and finally sintering it into compact and dense cemented carbide bodies.
[0103] WC powders of various particle sizes can be used as starting materials for the production of the powder mixture, in particular coarse-grained WC having a particle size FSSS>25 μm. Starting powders for the binder phase are extra-fine cobalt powder (FSSS 1.3 μm) and nickel aluminide, preferably nickel aluminide powder, in particular Ni.sub.3Al powder.
[0104] For instance, nickel aluminide powder (Ni—Al powder), for instance Ni-13Al powder with an aluminum content of about 13.3 wt % can be used. The particle size of the Ni—Al powder is FSSS<70 μm, preferably FSSS<45 μm. W metal powder (FSSS<2 μm) and lamp black are used to set and adjust a targeted carbon content. For alloying the binder phase with alloying elements, such as Ti, Ta, Mo, Nb, V, Cr, their carbide powders, or their W-containing mixed carbides having particle sizes <3 μm are used.

[0105] The powder mixture is produced according to the state of the art by wet grinding, preferably in a ball mill equipped with hard metal balls. Ethanol and hexane are used as grinding media. Other possible grinding media would be acetone or aqueous media with suitable inhibitors.
[0106] In the production of powder mixtures for cemented carbide material having binder contents >15%, a single grinding process is sufficient due to the high binder content and favored recrystallization. For binder contents up to 15%, on the other hand, a multi-stage wet grinding process is advantageous in order to effectively comminute the Ni—Al powders and to minimize the formation of oxides during the grinding process.
[0107] In the first step, the Ni—Al powder is intensively mixed with grinding fluid and coarse-grained tungsten carbide having a mean particle size FSSS>20 μm, preferably from 30 to 60 μm. If necessary, pressing aids, small quantities of alloying constituents and cobalt powder can also be added at this stage.
[0108] The grinding parameters (duration, ratio of grinding balls to grinding stock, grinding medium) and the ratio of WC to Ni—Al powder are based on the WC grain size to be set in the cemented carbide material.
[0109] In the second step, 50 to 80 wt % WC raw material(s) of defined particle size(s) is/are added at this pre-grinding stage and blended, wherein the main focus is on reducing agglomerates and obtaining as homogeneous a mixture as possible.
[0110] If the alloy adjustment and the addition of pressing aids were not performed in the first grinding step (pre-grinding stage VM), it can now be done in the second step.
[0111] The slurry obtained during wet grinding is dried according to the state of the art and converted into a powder ready for pressing. Preferably, this is done using the process of spray drying.
[0112] Forming is preferably performed directly, by axial pressing using mechanical, hydraulic or electromechanical presses.
[0113] Sintering is performed between 1350 and 1550° C. in a vacuum, preferably in industrial sintering HIP furnaces, in which an inert gas inlet creates overpressure after liquid phase sintering, wherein any residual porosity can be eliminated.
[0114] By way of example, FIG. **4** shows the WC—Co—Ni.sub.3Al phase diagram for 3 wt % Co and 3 wt % Ni.sub.3Al, which illustrates the formation of these precipitates.
[0115] After solidification of the melt, initially only WC and a solid solution of Co, Ni, Al, W and C are present. Only below the solvus temperature does the intermetallic phase material precipitate from this solid solution, wherein the intermetallic phase material is formed according to the structural formula (M,Y).sub.3(Al,X), wherein M=Ni, Y=Co and/or another constituent and X=tungsten and/or another constituent. A scanning electron microscope can be used to visualize these intermetallic phase materials.
[0116] FIGS. **2** and **3** illustrate two different cemented carbide materials according to the invention, in the form of hard metals, using such scanning electron micrographs. The binder phase of such a hard metal can be clearly seen, in which the intermetallic phase material (lighter phase) **10** and the metallic binder material **30** (dark) can be identified. The WC grains **20** are bonded by the binder phase.
[0117] A uniform distribution of the intermetallic phase material in the binder phase is shown, wherein the crystals of the intermetallic phase material have a cubic shape and are preferably smaller than 1500 nm. The crystals of the intermetallic phase material (M, Y).sub.3(Al, X) have a crystal structure L12 (space group 221) in accordance with ICSD (Inorganic Crystal Structure Database).
[0118] Such a cemented carbide material can be bonded to a steel base body to form a working element of a tool, for instance a crushing, soil working tool, preferably for a road milling machine, a recycler, a stabilizer, an agricultural or silvicultural soil cultivation machine. This working element is then disposed in the working area of the tool. The connection to the base body is made

using a soldered or brazed joint, in particular a brazed joint. In this process, heat is introduced into the tool to create the soldered or brazed joint. The tool is then quenched, for instance in a water-oil emulsion.

[0119] During the soldering or brazing process, the intermetallic phase material at least partially dissolves again, such that constituents of the intermetallic phase material are present as dissolved constituents in the cemented carbide material after quench hardening. In this way, a precursor cemented carbide material is formed.

[0120] This precursor cemented carbide material is then subjected to thermal treatment, as described several times above. In this regard, heat may be introduced into the cemented carbide material via the thermal treatment, wherein the temperature is below the solvus temperature and preferably above 400° C. The treatment duration, i.e., the time in which the thermal treatment takes place, is in the range from 0.25 to 24 h. During thermal treatment, intermetallic phase material forms again, at least in portions of the cemented carbide material, to effect a reinforcement of the binder phase.

[0121] The thermal treatment can be an active process, in which heat is selectively introduced into the cemented carbide material by means of a heat source. Preferably, the heat treatment is performed passively, wherein, for instance during tool application, the precursor cemented carbide material comes into contact with the workpiece to be machined, for instance a road pavement. During this contact, heat is introduced into the precursor cemented carbide material, i.e., it is heated to a temperature at which the intermetallic phase material forms. In this way, the tool automatically reinforces itself according to the invention, wherein the cemented carbide material according to the invention is formed in the area subject to wear.

[0122] It is also conceivable that a cemented carbide material designed in the manner described above is manufactured in a sintering process, in which intermetallic phases are formed. Subsequently, this product can be brought to a temperature, preferably above the solvus temperature, at which the intermetallic phase material at least partially redissolves. Then this material is quenched to form the precursor cemented carbide material. Then the precursor cemented carbide material is subjected to heat treatment to form the cemented carbide material of the invention.

[0123] In order to be able to easily precipitate the intermetallic phase material in the binder phase, provision may preferably be made for the (M,Y).sub.3(Al,X) content in the binder phase to be >40% and for the carbon balance to be set stoichiometrically or sub-stoichiometrically in that way.

[0124] It has been shown that a higher tungsten solution in the binder stabilizes the precipitation of the intermetallic phase material. This is caused by the incorporation of “Co3W” into the crystal structure of the intermetallic phase material and the shift of the precipitation range towards higher temperatures.

[0125] The elements Mo, Nb, Cr, V and in particular Ti, Ta, which can be added in small amounts (<15 at % in the binder) show a similar effect.

[0126] The usable alloy quantity depends on the individual solubility product of the metal carbides. Even though these appear negligible in terms of their magnitude, surprisingly clear effects are evident that cannot be attributed to a grain-reducing effect.

[0127] Due to the increased stability and better precipitation behavior, through the addition of further elements, the proportion of intermetallic phase material in the binder can be reduced and can also be lower than 40%. Furthermore, in the presence of, for instance, Ti or Ta, the carbon balance no longer needs to be set substoichiometrically because these elements take over the role of tungsten as stabilizer.

[0128] The effect of precipitation of the intermetallic phase material on high-temperature strength can be impressively demonstrated by hot compression tests. FIG. **1** shows the hot compressive strength of hard metals containing 6% binder each at different test temperatures and strain rates. In particular, the intermetallic phase material increases the strength by approx. 40 to 50% at a test

temperature of 800° C.
[0129] Physical quantities are determined on the cemented carbide material samples according to the invention, which contribute to characterize the material and its properties.
[0130] For hard metals, the determination of the coercivity H.sub.cM and the specific magnetic saturation 4 ps have been established as non-destructive testing methods.
[0131] Both measured variables are also determined for the characterization of the cemented carbide material of the invention using a Koerzimat® 1.097 by Förster.
[0132] Another parameter for characterizing the material is density, which is determined by weighing according to Archimedes' principle.
[0133] The hardness of the material is determined in accordance with the standard applicable to hard metals on metallographically prepared polished specimens. Preferably, the Vickers HV10 hardness test with a test load of 10 kp is used (ISO 3878).
[0134] Also, the porosity of the sintered material (EN ISO 4499-4 standard) and aluminum oxide particles are detected and evaluated by light microscopy on polished specimens. To estimate the volume percentages of aluminum oxide in the microstructure, comparative images of A porosity and B porosity can be used, wherein A08 and B08 are approximately equal to a volume fraction of 0.6 vol %. The Eta phase is etched with Murakami solution according to the standard (EN ISO 4499-4) for a light microscopic examination. The average WC grain sizes are determined according to DIN ISO 4499-2. In so doing, SEM (scanning electron microscope) images are evaluated using the linear-intercept technique.
[0135] The proportions of the intermetallic phase in the binder and the maximum size of the precipitated particles are also determined by SEM images, but using an inlense BSE detector. For this purpose, images are taken at several locations of the sample and the evaluation is performed on a representative section by means of image processing and determination of the area fractions by tonality demarcation.

Description

EXAMPLES
[0136] The table below shows examples of cemented carbide bodies according to the invention. The examples shown in this table can in principle be manufactured using the same method as described above:
TABLE-US-00001 Example Description 1 6% binder, thereof approx. 50% intermetallic phase, raw materials without carbon correction 2 6% binder, thereof approx. 50% intermetallic phase, addition of tungsten metal powder 3 6% binder, thereof approx. 50% intermetallic phase, addition of carbon black 4 6% binder, thereof approx. 40% intermetallic phase, raw materials without carbon correction 5 6% binder, thereof approx. 50% intermetallic phase, addition of WTiC 6 8.5% binder, thereof approx. 40% intermetallic phase, raw materials without carbon correction 7 15% binder, thereof approx. 50% intermetallic phase, raw materials without carbon correction 8 Small series containing 6% binder, thereof approx. 50% intermetallic phase, addition of tungsten metal powder Reference, not according 6% binder, cobalt only, grinding conditions similar to the invention to Examples 1 to 8 6-50 6-50 C− 6-50 C+ 6-40 Designation Example 1 Example 2 Example 3 Example 4 Powder Pre-grinding Material Size [kg] [kg] [kg] [kg] preparation Weighted sample Ni-13Al −325 0.120 0.120 0.120 0.096 mesh Co FSSS 0.120 0.120 0.120 0.144 1.3 µm WC FSSS 0.760 0.760 0.760 0.760 25 µm WTiC 50:50 FSSS 1.7 µm Total 1 1 1 1 Grinding Grinding time [h] 24 24 24 24 parameters Ratio of 5:1 5:1 5:1 5:1 grinding balls to grinding stock Grinding result Material [kg] [kg] [kg] [kg] Weighted sample Pre-grinding 0.25 0.25 0.25 0.25 Co FSSS 1.3 µm WC FSSS 0.75 0.745 0.75 0.75 25 µm W-Metal FSSS 0.005 2 µm Carbon black 0.0003 Total 1 1 1 1 Grinding Grinding time [h] 8 8 8 8 parameters Ratio of 5:1 5:1 5:1 5:1 grinding balls to grinding

stock Sintering Parameter(s) Sinter HIP Vacuum Furnace System: Vacuum solvent dewaxing, vacuum sintering/argon partial pressure, sintering temperature 1430° C. 1 h + 30′ high pressure 50 bar, pressurized cooling, cooling time in temperature interval 900-600° approx. 40 min physical coercive force H.sub.cM [kA/m] 21.9 25.5 15.1 11.5 characteristics spec. magn. 4πσ [μTm.sup.3/kg] 5.8 5.1 6.6 7.1 Saturation Hardness HV10 1200 1220 1170 1160 Density ρ [g/cm.sup.3] 14.83 14.87 14.82 14.85 Porosity EN ISO <A02, <A02, <A02, <A02, 4499-4 B00, C00 B00, C00 B00, C00 B00, C00 Grain size WC EN ISO 3 2.9 3.1 3.2 4499-2 Hot compressive [MPa] strength* Other Eta phase none none none none characteristics Alumina none none none none Binder content in the hard metal [m %] 6 6 6 6 (weighted sample) Proportion intermet. [%] 46% 48% 39% 35% Phase to Binder ** max. size of the [nm] <150 <150 <150 <150 intermetallic phase 6-50 Ti 8.5-40 15-50 6-50 C− S 6-0 Designation Example 5 Example 6 Example 7 Example 8 Reference Powder Pre-grinding Material Size [kg] [kg] [kg] [kg] [kg] preparation Weighted sample Ni-13Al −325 0.120 0.240 0.240 9.000 mesh Co FSSS 0.120 9.000 0.240 1.3 μm WC FSSS 0.740 0.760 0.760 57.000 0.760 25 μm WTiC 50:50 FSSS 0.020 1.7 μm Total 1 1 1 75 1 Grinding Grinding time [h] 24 24 24 7 24 parameters Ratio of grinding balls 5:1 5:1 5:1 6.7:1 5:1 to grinding stock Grinding result Material [kg] [kg] [kg] [kg] [kg] Weighted sample Pre-grinding 0.25 0.142 0.313 37.5 0.25 Co FSSS 0.051 0.075 1.3 μm WC FSSS 0.75 0.807 0.612 111.7 0.75 25 μm W-Metal FSSS 0.8 2 μm Carbon black Total 1 1 1 150 1 Grinding Grinding time [h] 8 8 8 6 10 parameters Ratio of grinding balls 5:1 5:1 5:1 3.3:1 5:1 to grinding stock Sintering Parameter(s) Sinter HIP Vacuum Furnace System: Vacuum solvent dewaxing, vacuum sintering/argon partial pressure, sintering temperature 1430° C. 1 h + 30′ high pressure 50 bar, pressurized cooling, cooling time in temperature interval 900-600° approx. 40 min physical coercive force H.sub.cM [kA/m] 25.3 11.7 20.4 28.7 5.2 characteristics spec. magn. 4πσ [μTm.sup.3/kg] 5.4 10.0 13.9 4.9 11.5 Saturation Hardness HV10 1205 1080 910 1250 1150 Density ρ [g/cm.sup.3] 14.83 14.53 13.75 14.82 14.91 Porosity EN ISO <A02, <A02, <A02, <A02, <A02, 4499-4 B00, C00 B00, C00 B00, C00 B00, C00 B00, C00 Grain size WC EN ISO 2.9 3 3.1 2.4 3.1 4499-2 Hot compressive [MPa] 1930 1380 strength* Other Eta phase none none none none characteristics Alumina none none <0.10 vol. % <0.05 vol. % Binder content in the hard metal [m %] 6 8.5 15 6 6 (weighted sample) Proportion intermet. [%] 49% 35% 47% 52% — Phase to Binder ** max. size of the intermetallic phase [nm] <150 <150 <150 <100 — *non-standardized comparative test using specimens ø8 × 12 mm, test temperature 800° C., strain rate 0.001 1/s ** Evaluation of area proportions based on tonality demarcation in the micrograph. Calibration using solution-annealed samples isothermally aged at 700°/10 h of the same composition.

[0137] According to the above, the invention thus relates to a cemented carbide material, in particular hard metal, containing 70 to 95 wt %, preferably containing 80 to 95 wt %, tungsten carbide in dispersed form, and a binder phase, wherein the binder phase comprises metallic binder material, wherein the metallic binder material comprises Co, dissolved Ni and dissolved Al, wherein the binder phase may optionally comprise intermetallic phase material, wherein the intermetallic phase material, if present, is formed according to the structural formula (M,Y).sub.3(Al,X), wherein M=Ni, Y=Co and/or another constituent and X=tungsten and/or another constituent, wherein the binder phase has the chemical element composition listed below: Ni>25 wt %, Al>4 wt %, the balance is made up of Co and dissolved binder constituents, for instance W and/or C.

[0138] The invention is explained in greater detail below based on the exemplary embodiments shown in FIGS. **5** to **12**. In the figures,

[0139] FIG. **5** shows a vertical section through a pick tip,

[0140] FIG. **6** shows the pick tip according to FIG. **5** along the course of the section line marked VI-VI in FIG. **5**, and

[0141] FIGS. **7** to **12** show a vertical section of the pick tip **50** of FIGS. **5** to **6**, but with a modified microstructure composition.

[0142] FIGS. **5** to **12** show pick tips **50** in the form of a cemented carbide material **40**. Advantageously, these milling picks **50** are for use at a cutting tool, in particular a cutting pick, a round pick, a road milling pick, a mining pick or the like.

[0143] The pick tip **50** is designed and prepared to be connected, preferably soldered or brazed, to a steel body. For this purpose, the steel body usually has a head to which a shank, preferably a round shank, is integrally formed. Facing away from the shank, the head has a mount for the pick tip **50**. The pick tip **50** can be fastened in or to this mount.

[0144] The pick tip **50** is integrally designed and has a base part **51**. The base part **51** can be used to connect the pick tip **50** to the steel body. Preferably, the base part **51** has a connection surface **51.1**. For attaching the pick tip **50** to the steel body, the brazing material of a brazed joint may be disposed between the connection surface **51.1** and the steel body.

[0145] To maintain as constant a thickness as possible of the brazing gap between the pick tip **50** and the steel body, provision may be made for spacers **51.2** to be formed on the pick tip **50** in the area of the connection surface **51.1**, which project beyond the connection surface **51.1** and are designed to rest on the steel body in such a way that the connection surface **51.1** is kept at a distance from a mating surface of the steel body for the soldering or brazing process.

[0146] Furthermore, provision may be made for one or more recesses **52** to be present in the area of the connection surface **51.1**. In this case, the recess can preferably be such that it merges from the connection surface **51.1** via a convex fillet into a recessed section, which is advantageously designed as a concave trough. The recess **52** can be used to reduce the amount of material required for the pick tip **50**. In addition, the recess **52** forms a reservoir for excess solder or brazing material in the area of the connection surface **51.1**.

[0147] The base part **51** has a preferably circumferential edge **51.3**, which can be formed as an at least sectionally convex formation, the edge **51.3** can be formed as a transition between the base part **51** and a transition section **53**.

[0148] The transition section **53** has a first segment formed as a concave segment **53.1**. Alternatively, a truncated cone-shaped geometry or a combination consisting of a concave segment **53.1** and an at least sectionally truncated cone-shaped geometry can also be provided. In the first segment, the pick tip **50** is tapered in the direction from the base part **51** toward a tip **54** of the pick tip **50**.

[0149] Further, the transition section **53** may also include a cylindrical segment **53.2** that adjoins the first segment opposite from the base part **51**. Preferably, the transition, at least in areas of the pick tip **50**, between the first segment and the cylindrical segment is continuous, preferably continuously differentiable in the direction of the central longitudinal axis of the pick tip, such that jumps in continuity are avoided, as shown in FIG. **5**.

[0150] The pick tip **50** can preferably have indentations **53.3** in the area of the transition section **53**. They are used to reduce material and optimize the drainage of soil material that is removed during the application of tools.

[0151] The pick tip **50** has a tip **54** that adjoins the transition section **53**, preferably the cylindrical segment **53.2**. The connection section **54.1** can be designed as a convex curvature. A taper section **54.2**, which merges into an end section **54.3** adjoins the connection section **54.1**. The end section **54.3** is, preferably in the form of a convex curvature, particularly preferably in the form of a spherical cap.

[0152] FIG. **6** shows a cross-section and a top view of the pick tip **50**. The indentations **53.3** are evenly distributed around the circumference of the pick tip **50** as can be clearly seen.

[0153] The pick tip **50** has least one first volume segment **70** and least one second volume segment **60**.

[0154] In the second volume segment **60**, the cemented carbide material **40** has tungsten carbide (WC) grains **20**, which are bonded to each other via metallic binder material **30** of a binder phase. Provision may be made for no intermetallic phase material **10** or intermetallic phase material **10** preferably at a concentration of less than 30 wt %/unit volume, preferably less than 25 wt %/unit volume, particularly preferably less than 15 wt %/unit volume to be present in the second volume segment **60**.

[0155] In the first volume segment **70**, the cemented carbide material **40** comprises tungsten carbide (WC) grains **20** bonded via metallic binder material **30** of a binder phase. In the first volume segment **70**, intermetallic phase material **10** is present, preferably at a concentration >30 wt %/unit volume, preferably in the range from 30 to 70 wt %/unit volume, particularly preferably in the range from 35 to 60 wt %/unit volume, further preferably in the range from 40 to 50 wt %/unit volume.

[0156] The intermetallic phase material is formed according to the structural formula (M,Y).sub.3(Al,X), wherein M=Ni, Y=Co and/or another constituent and X=tungsten and/or another constituent. The binder phase has the chemical element composition listed below: Ni>25 wt %, Al>4 wt %, the balance is made up of Co and dissolved binder constituents, for instance W and/or C.

[0157] The relative proportion Y, in relation to a unit volume, of intermetallic phase material in the first volume segment **70** is greater than in the second volume segment **60**.

[0158] As shown in the drawings, the first volume segment **70** and the second volume segment **60** may be delimited by at least a part of the surface of the cemented carbide material **40**.

[0159] The first volume segment **70**, which has the relatively higher proportion of intermetallic phase material **10**, forms part of the surface of the pick tip **50**, in particular at the tip **54** at the taper section **54.2** and/or at the end section **54.3**.

[0160] Starting from the tip **54**, the first volume segment **70** may further extend up to the base part **51**. Preferably, the area of the base part **51** can also be formed by the second volume segment **60**. The second volume segment **60** with the relatively lower content of intermetallic phase material **10** or without intermetallic phase material **10** is preferably disposed in the area of the transition section **53**. As FIG. **6** illustrates, both the first volume segment **70** and the second volume segment **60** may be adjacent to the surface of the pick tip **50** in the area of the transition section **53**.

[0161] The arrangement of the first and second volume segments **70**, **60** according to the embodiment example shown in FIGS. **5** and **6** has the technical advantages listed below:

[0162] Because the first volume segment **70** forms the tip **54**, a high abrasion resistance is provided here in the area that is particularly susceptible to wear.

[0163] In the non-wear area, this results in increased toughness and thus higher fracture stability.

[0164] Furthermore, it is conceivable that the volume segments can be used to set **60**, **70** specific wear shapes, which, for instance, support re-sharpening of the pick tip.

[0165] FIGS. **7** to **12** show further design variants of a pick tip **50**. To this end, the pick tip **50** is designed largely identically to the pick tip **50** of FIGS. **5** and **6**. In this respect, to avoid repetition, reference is made to the above statements and only the differences are discussed. The pick tips **50** according to FIGS. **7** to **12** differ in particular in the arrangement and design of the first and second volume segments **70**, **60**.

[0166] As FIG. **7** illustrates, the first volume segment **70** is located in the area of the tip **54** and preferably partially in the cylindrical area **53.2** of the transition section **53**. However, it is also conceivable that the first volume segment **70** is disposed only in the area of the tip **54**. There, the first volume segment **70** effectively protects against abrasive wear in the area of the tip **54**.

[0167] FIG. **8** illustrates that the first volume segment **70** may be located entirely or partially inside the pick tip **50**. The first volume segment **70** can be designed in such a way that it preferably extends in the direction of the central longitudinal axis of the pick tip **50** across the entire area of the transition section **53**. As mentioned above, the first volume segment **70** is characterized by a particularly high shear strength as a result of the presence of intermetallic phase material **10**. In this way, the first volume segment **70** effectively reinforces the fracture-prone transition area **53**.

[0168] Conversely, as FIG. **9** shows, the second volume segment **60** may be located wholly or

partially inside the pick tip **50**. The first volume segment **70** then preferably completely encompasses the second volume segment **60**. The first volume segment completely or almost completely forms the surface of the pick tip **50**, to protect this tip in particular effectively against abrasion and to prevent the pick tip **50** from breaking in the area of the transition section **53**. Owing to the large extension transverse to the central longitudinal axis of the pick tip **50**, the first volume segment **70** also has a high equatorial modulus of resistance against bending.
[0169] FIG. **10** illustrates that, in further development of the variant of FIG. **7**, the first volume segment **70** with its relatively high proportion of intermetallic phase material **10** can extend across the area of the tip **54** and the transition section **53**, such that it preferably completely forms the surface of the tip **54** and the transition section. The first volume segment **70** is extended up to the base part **51** or is also extended into the base part **51**, as illustrated in FIG. **10**.
[0170] FIG. **11** shows that the first volume segment **70** may also extend inside the pick tip **50** to form a continuous volume segment, from the end section **54 3** of the tip **54** to the base part **51**.
[0171] FIG. **12** illustrates that, in reversal to the embodiment example according to FIGS. **5** and **6**, provision may also be made for the first and second volume segments **70**, **60** to be interchanged.
[0172] The above drawings shall thus be understood to denote schematic representations. In particular, the first and second volume segments **70**, **60** do not form exactly sharply defined areas as shown in the drawings, but rather transition sections form between the two volume segments **70**, **60**.
[0173] For the production or formation of the cemented carbide materials **40** shown in FIGS. **5** to **12**, a method is used according to the invention, wherein first in a first process step a precursor cemented carbide material, in particular a hard metal, is created, which contains 70 to 95 wt %, preferably 80 to 95 wt %, tungsten carbide in dispersed form, and a binder phase. The binder phase has at least Co as the metallic binder material and the dissolved elements Ni and Al. The binder phase has the chemical element composition listed below:
[0174] Ni>25 wt %, Al>4 wt %, the balance is made up of Co and dissolved binder constituents, for instance W and/or C.
[0175] In a further process step, the precursor cemented carbide material is subjected to heat treatment as explained above to form the cemented carbide material **40**, which has intermetallic phase material **10** in the binder phase at least in the first volume segment **70**. The intermetallic phase material **10** is formed according to the structural formula (M,Y).sub.3(Al,X), wherein M=Ni, Y=Co and/or another constituent and X=tungsten and/or another constituent.
[0176] To this end, in the further process step, the precursor cemented carbide material may be maintained in a temperature range between 400° C. and the solvus temperature during heat treatment for a time period ranging from 0.25 to 24 hours. For the targeted formation of the individual volume segments **60**, **70**, for instance, targeted heating can be performed by means of a laser or an induction coil.

Claims

1-22. (canceled)
23. A method for producing a cemented carbide body, comprising: combining a tungsten carbide powder and a metallic binder material comprising cobalt, nickel, and aluminum to form a powder mixture; adding nickel aluminide to the powder mixture; forming a green compact from the powder mixture; sintering the green compact at a first temperature and a first pressure; and cooling the compact to form a cemented carbide body comprising a binder phase and an intermetallic phase.
24. The method of claim 23, wherein the tungsten carbide powder and the metallic binder material are combined in a mixing process or a milling process.
25. The method of claim 24, wherein the mixing process or the milling process is a multi-stage process comprising at least a first mixing or milling step and a second mixing or milling step, and

wherein the nickel aluminide is added before a final mixing or milling step.
26. The method of claim 24, wherein the tungsten carbide powder and the metallic binder material are combined in a wet milling process.
27. The method of claim 23, wherein at least a portion of the intermetallic phase comprises (M,Y).sub.3(Al,X), wherein M is Ni, Y comprises Co, and X comprises W.
28. The method of claim 23, wherein, when cooling the compact, the sintered body is kept at a temperature range between 400° C. and a solvus temperature of the sintered body for a period ranging from 0.25 to 24 hours.
29. The method of claim 23, wherein the green compact comprises tungsten carbide in an amount ranging from 70 wt % to 95 wt % of the green compact, cobalt in an amount ranging from 1 wt % to 19 wt % of the green compact, and nickel aluminide in an amount ranging from 1 wt % to 28 wt % of the green compact.
30. The method of claim 23, wherein the nickel aluminide is produced in a smelting process or added as a material produced in a melt process.
31. The method of claim 23, wherein the nickel aluminide has a mean particle size of less than 70 μm.
32. The method of claim 23, further comprising a preparation step, the preparation step comprising combining the nickel aluminide with a milling liquid and coarse-grained tungsten carbide to form a milling mixture and mixing the milling mixture such that crushed nickel aluminide is formed from the nickel aluminide.
33. The method of claim 32, wherein the tungsten carbide has a mean particle size of greater than 20 μm.
34. The method of claim 32, wherein the tungsten carbide is macrocrystalline tungsten carbide, monocrystalline tungsten carbide, or a combination thereof.
35. The method of claim 32, wherein the preparation step or a subsequent milling step comprises mixing at least the cobalt, at least one alloying constituent, or a combination thereof with the nickel aluminide, the crushed nickel aluminide, or a combination thereof.
36. The method of claim 32, wherein the milling mixture comprises nickel aluminide in an amount ranging from 8 wt % to 50 wt %
37. The method of claim 32, further comprising a milling step subsequent to the preparation step, wherein tungsten carbide powder is added to the milling mixture until the milling mixture contains an amount of tungsten carbide ranging from 70 wt % to 95 wt %, and wherein the crushed nickel aluminide is ground into finely-crushed nickel aluminide.
38. The method of claim 23, wherein the green compact is sintered at a temperature of 1350° C. to 1550° C. using a liquid phase sintering process.
39. The method of claim 38, wherein the cobalt and the intermetallic phase are at least partially dissolved in each other in a melt during the liquid phase sintering process, wherein, when cooling the compact or during a thermal treatment step subsequent to the sintering process, the intermetallic phase material is formed in the binder phase, and wherein the intermetallic phase material has a structural formula of (M,Y).sub.3(Al,X), wherein M is Ni, Y comprises Co, and X comprises tungsten.
40. The method of claim 23, wherein at least a portion of the intermetallic phase material has a maximum particle size of 1500 nm and wherein at least a portion of the intermetallic phase material has an L12 crystal structure.
41. The method of claim 23, wherein the nickel aluminide is present as an intermetallic phase material.
42. The method of claim 23, further comprising adding Nb, Ti, Ta, Mo, V, Cr, or a combination thereof to the powder mixture, such that the metallic binder material further comprises the Nb, Ti, Ta, Mo, V, Cr or the combination thereof.
43. The method of claim 42, wherein the Nb, Ti, Ta, Mo, V, Cr or the combination thereof is

dissolved in the binder phase.
44. The method of claim 23, wherein a carbon content of the cemented carbide material is stoichiometric or substoichiometric, and wherein the carbon content in the cemented carbide material ranges from C.sub.stoich (wt %)-0.003*binder content (wt %) to C.sub.stoich (wt %)-0.012*binder content wt %.
45. The method of claim 23, wherein the binder phase contains 15 at % or less combined Nb, Ti, Ta, Mo, V, and Cr content.
46. The method of claim 23, wherein the tungsten carbide is present in the cemented carbide material as grains having a mean particle diameter ranging from 1 μm to 15 μm.
47. The method of claim 23, wherein the binder phase comprises less than 5 wt % Fe.